

TrES-3b

R-1711

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210417 · created 2026-08-02T15:32:36+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459321.8349 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.018
Transit depth (Rp/Rs)^2	2.81 +/- 0.59 [%]
Orbital Inclination (inc)	82.82 +/- 1.03
Airmass coefficient 1 (a1)	1.076 +/- 0.013
Airmass coefficient 2 (a2)	-0.0531 +/- 0.0063
Scatter in the residuals of the lightcurve fit is	1.18 %
Best Comparison Star	#2 - [26, 128]
Optimal Method	PSF photometry
Transit Duration (day)	0.0623 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.168 ± 0.018 vs 0.16309 (deviation 0.27σ)
Duration fit vs published	0.0623 d vs 0.0567 d (deviation 0.66σ)
Mid-time O–C	-1.8 min
Depth SNR	9.3
Residual scatter	1.18 %

Light curve

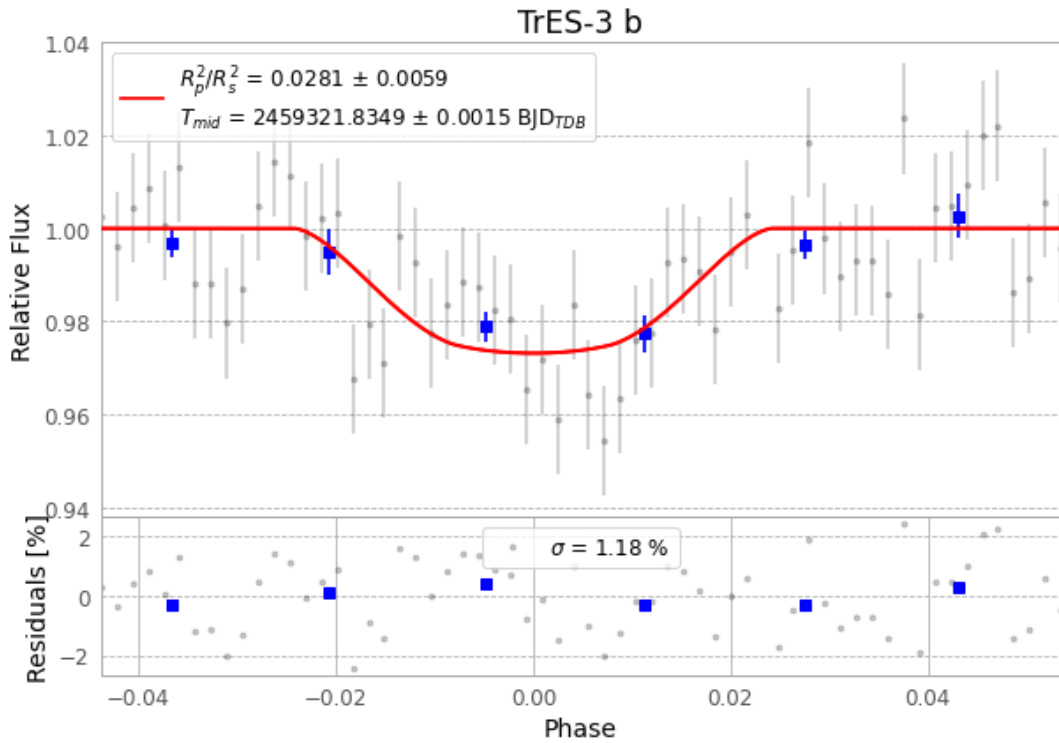


Figure 1 — FinalLightCurve_TrES-3 b_2021-04-16.png (SHA-256 c7151e3653c3c260...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [325, 255], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3ba37d1a02a4a1ae...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

61 FITS frames (40 MB), TRES-3210417063612.FITS → TRES-3210417093912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210417.log (0b4772fa6a6a...), FinalLightCurve_TrES-3 b_2021-04-16.png (c7151e3653c3...), FinalLightCurve_TrES-3 b_2021-04-16.csv (c1996a2d7ad7...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 15:32Z.